ST2187 Business Analytics, Applied Modelling and Prediction

Tableau Analysis

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Executive Summary

This analysis aims to evaluate market performances, profitability and operational problems with a focus on Southeast Asia's (SEA) Philippines. By assessing sales, profits, shipping delays and discount strategies, I have provided data-driven recommendations to improve business outcomes.

This report aims to answer the following questions:

- Which Regions/Countries have the highest and lowest profits?
- What are the issues or problems that the company is facing that causes these problems?
- What are some recommended actions the company can take to address these issues?

Findings indicated that high shipping delays and excessive discount strategies are major contributors to the low profits. By implementing the recommendations provided, the business would be able to reduce unnecessary discounts, improving customer relations and as a result, enhance profitability and efficiency of the business.

1. Introduction

The Southeast Asia (SEA) market presents significant business opportunities. By 2030, SEA is predicted to add about 140 million new consumers (WEF,2020). By that time, about 1 in 6 consuming household globally would be in SEA. However, there will also be operational challenges that impact profitability.

1.1. Objectives

This report will delve into the 5 dashboards created for the business that identifies its market trends, shipping efficiency and discount strategies to identify areas for improvement. The primary objectives of this analysis are:

- To access profit and sales of the business
- Conduct an analysis of SEA
- Conduct an analysis of Shipping and Discounts
- Recommend Solutions

2. Analysis

2.1. Dashboard 1: Market Analysis

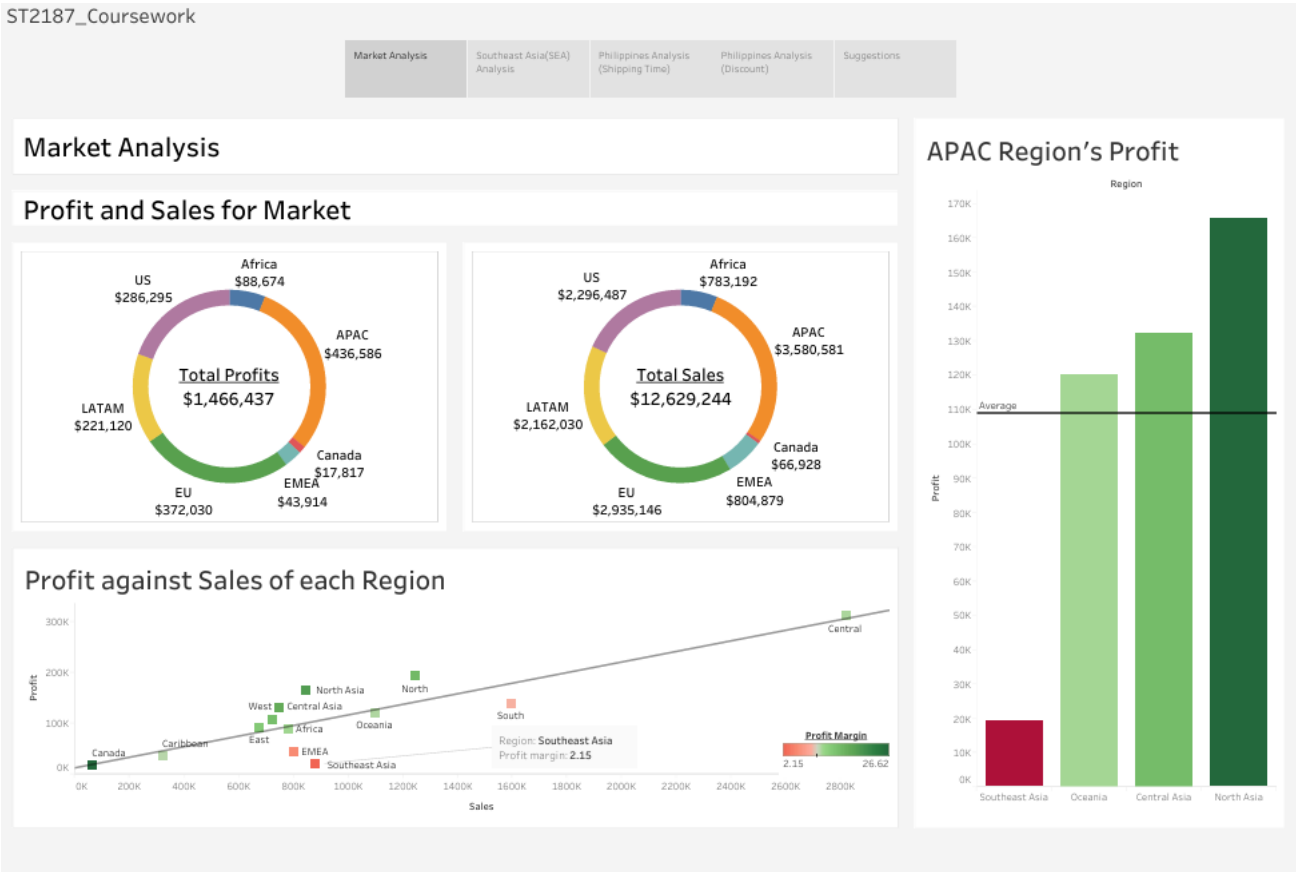

Fig.1 Market analysis for business

To further understand the business context, this market analysis was performed that includes overall business performance and performance of regions.

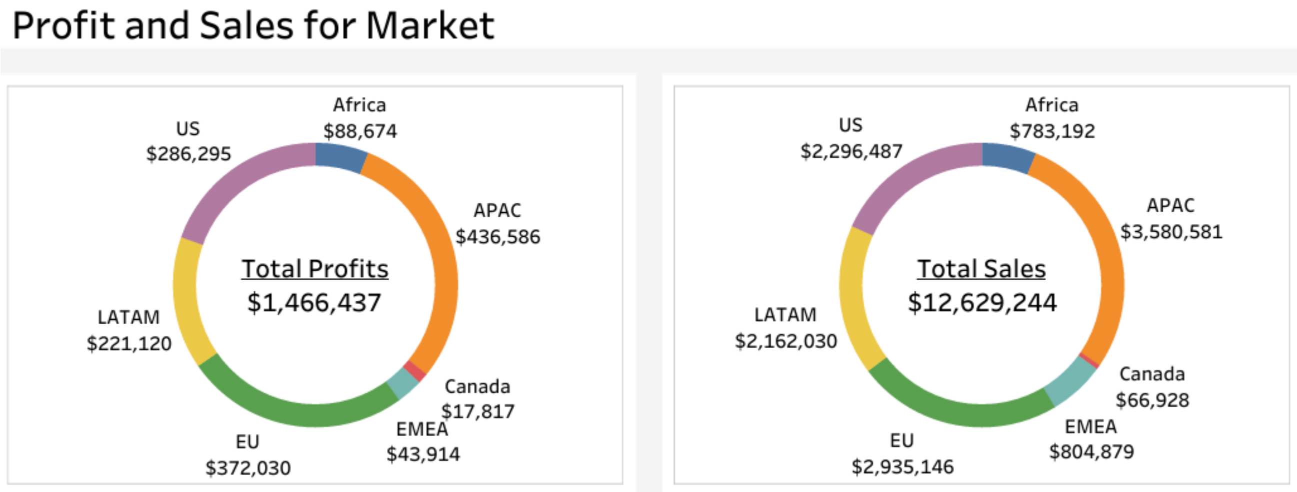

Fig 2. Profit and Sales figure of Business

The business had a total sale of $12,629,244 and of that, total profit was at $1,466,437. This leads to a profit margin of about 11.6%. While a profit margin of 11.6% can be deemed healthy (Investopedia, 2024), there is always room for improvement. The APAC market is contributing the highest profit and sales at $436,586 and $3,580,581.

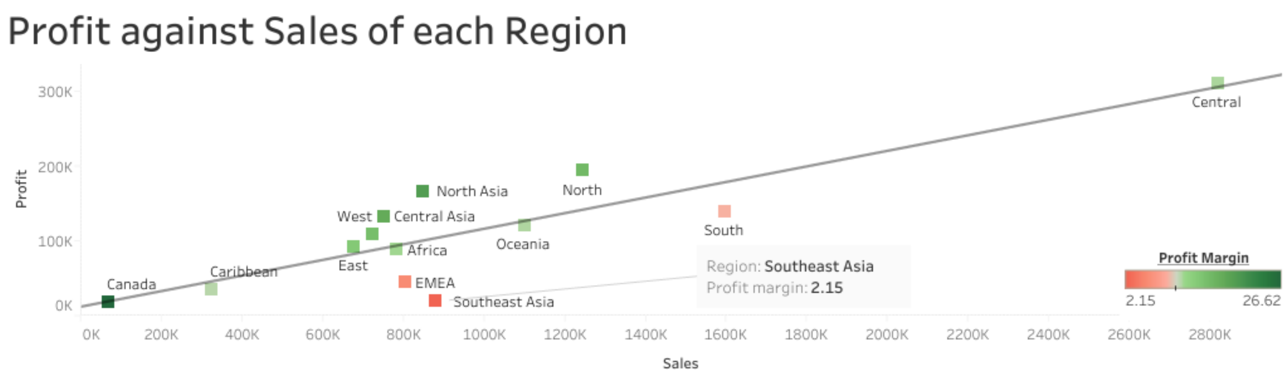

Fig 3. Profit and Sales based on region

The Profit against Sales scatterplot **(Figure 3)** provides a clear visual representation of SEA underperformance relative to other regions. With a profit margin of only 2.15%, SEA is significantly lagging.

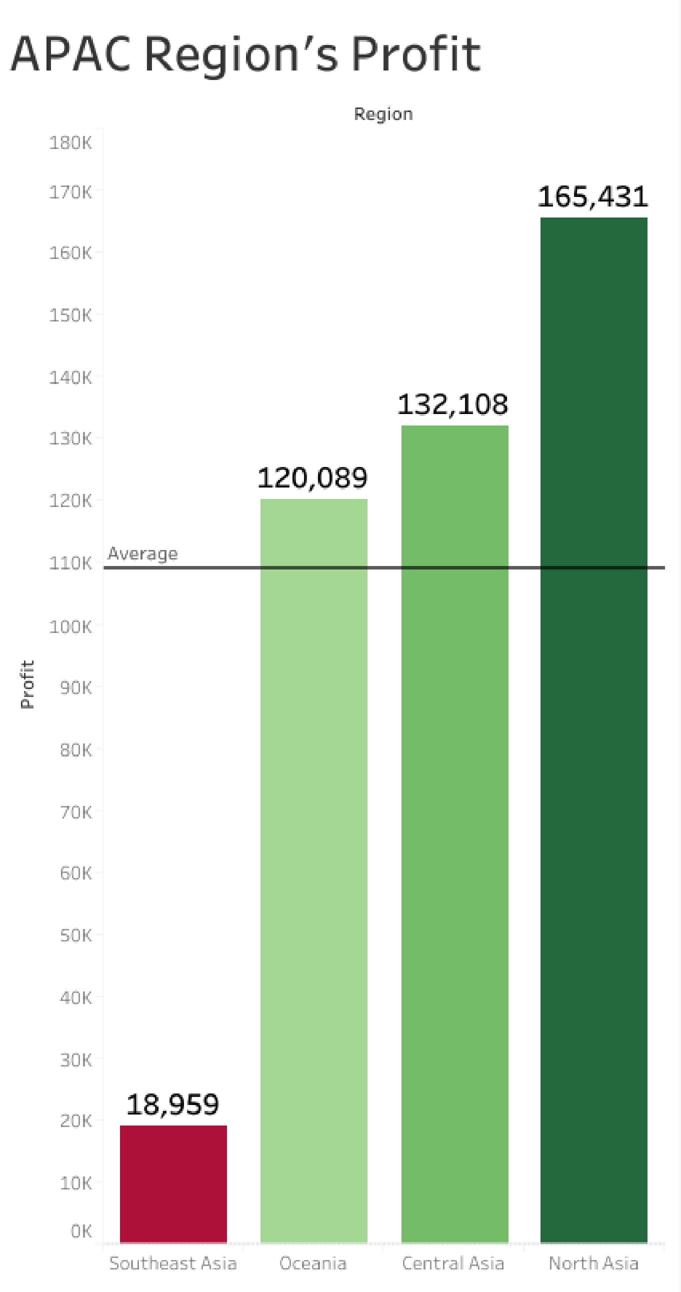

Fig. 4 APAC region's Profits

This shows that SEA has the lowest profits as compared to the other regions in APAC. This indicates potential inefficiencies or market challenges that requires further investigation, which we will do so in the subsequent dashboards.

2.2. Dashboard 2: Southeast Asia (SEA) Analysis

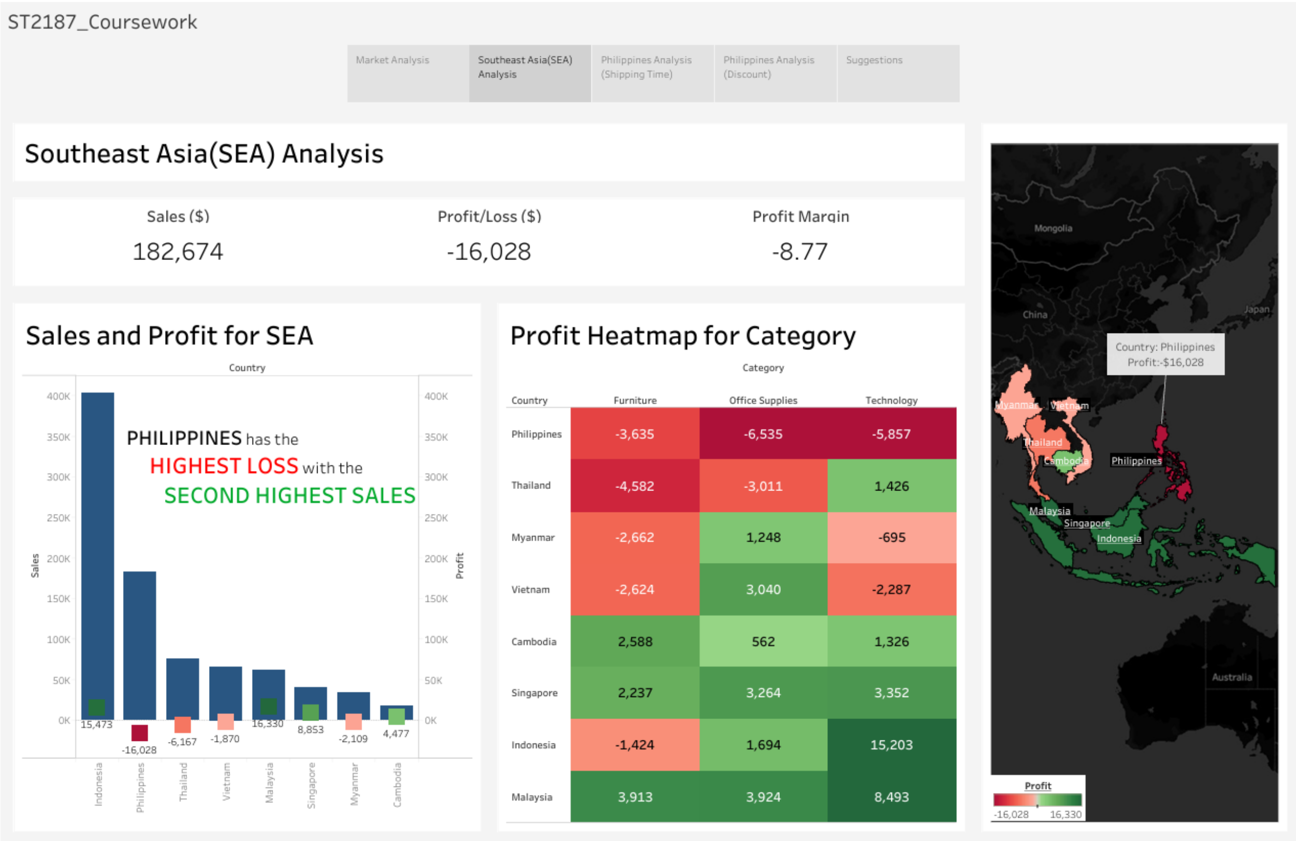

Fig 5. Southeast Asia (SEA) Analysis dashboard

To figure out what exactly is causing Southeast Asia to have such low profits, an analysis is carried out (Figure 5). Based on the dashboard, Southeast Asia have a total sale of $182,674, a loss of $16,028 which resulted in a negative profit margin of 8.77. This means that for every $100 in sales that they made, it will result in $8.77 in loss, highlighting financial inefficiencies for the region.

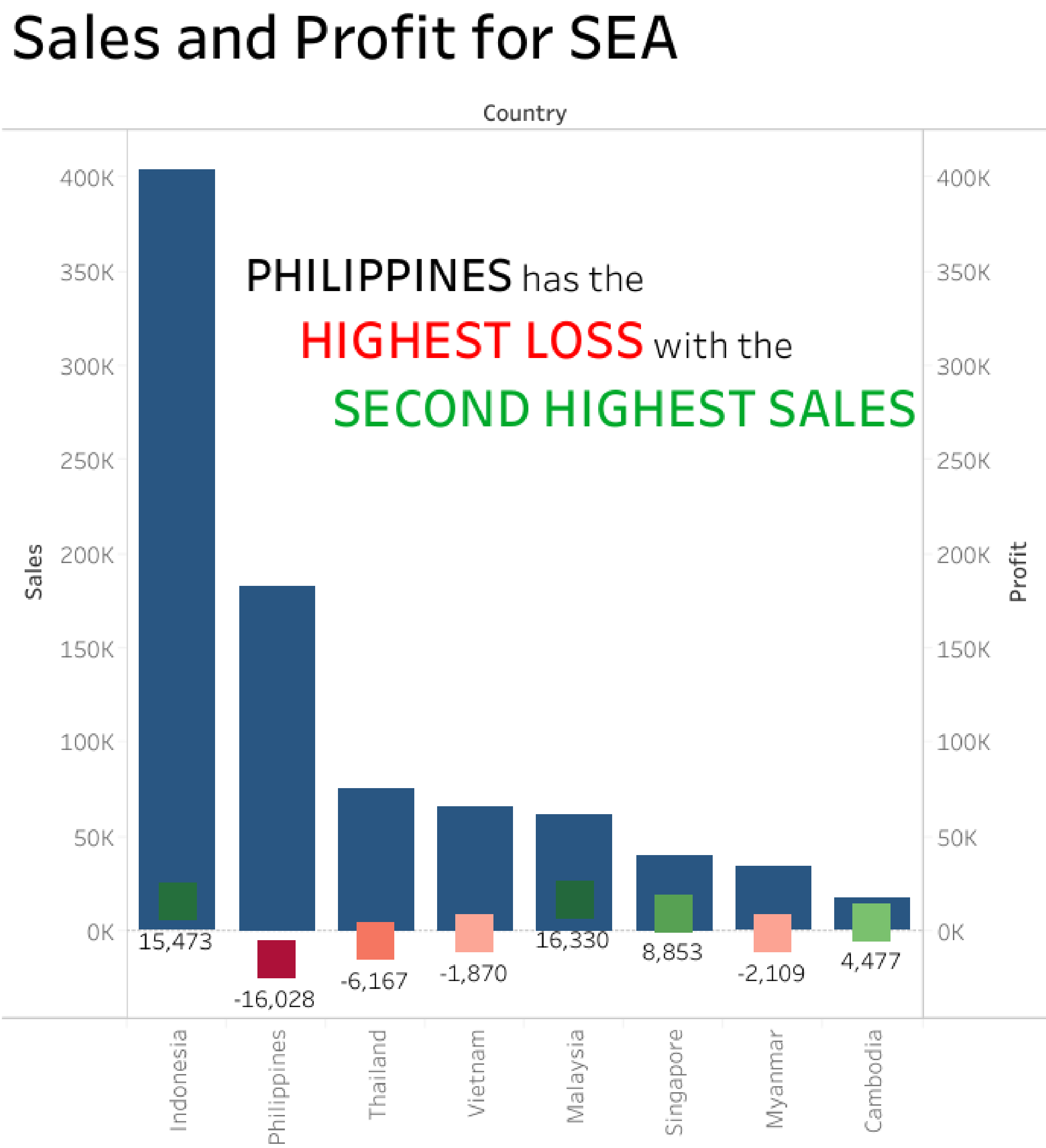

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Fig 6. Sales and Profit for SEA

In Southeast Asia, while the Philippines has the second highest sales, they also have the highest loss at $16,028. This is an important finding as it shows that despite Philippines being able to generate sales, high cost and other contributing factors are driving it into a loss.

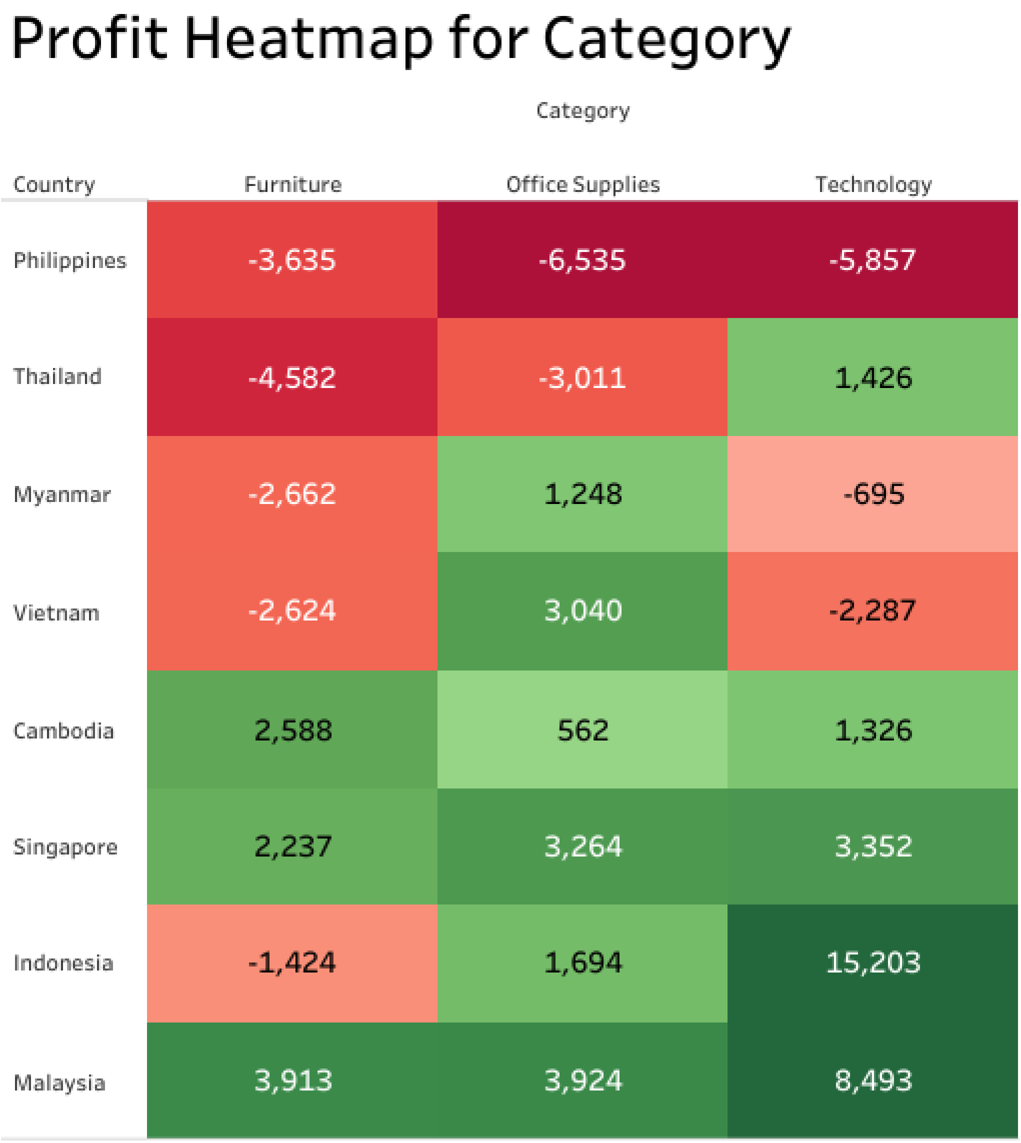


Fig 7. Profit heatmap based on Categories

This heatmap analysis shows that every product category in the Philippines is operating at a loss, with Office Supplies and Technology experiencing the highest loss compared to the rest. Philippines furniture category is also contributing the second highest loss. On the other hand, Malaysia, Singapore and Cambodia demonstrated consistent profitability across all categories. Meanwhile, Thailand, Myanmar, Vietnam and Indonesia show a mix of both profitable and unprofitable categories.

To better understand why Philippines is performing so badly, the next analysis will go deeper into shipping delays and discount strategies, both of which may be significantly affecting Philippines ability to generate profits.

2.3. Dashboard 3: Philippines Analysis (Shipping time)

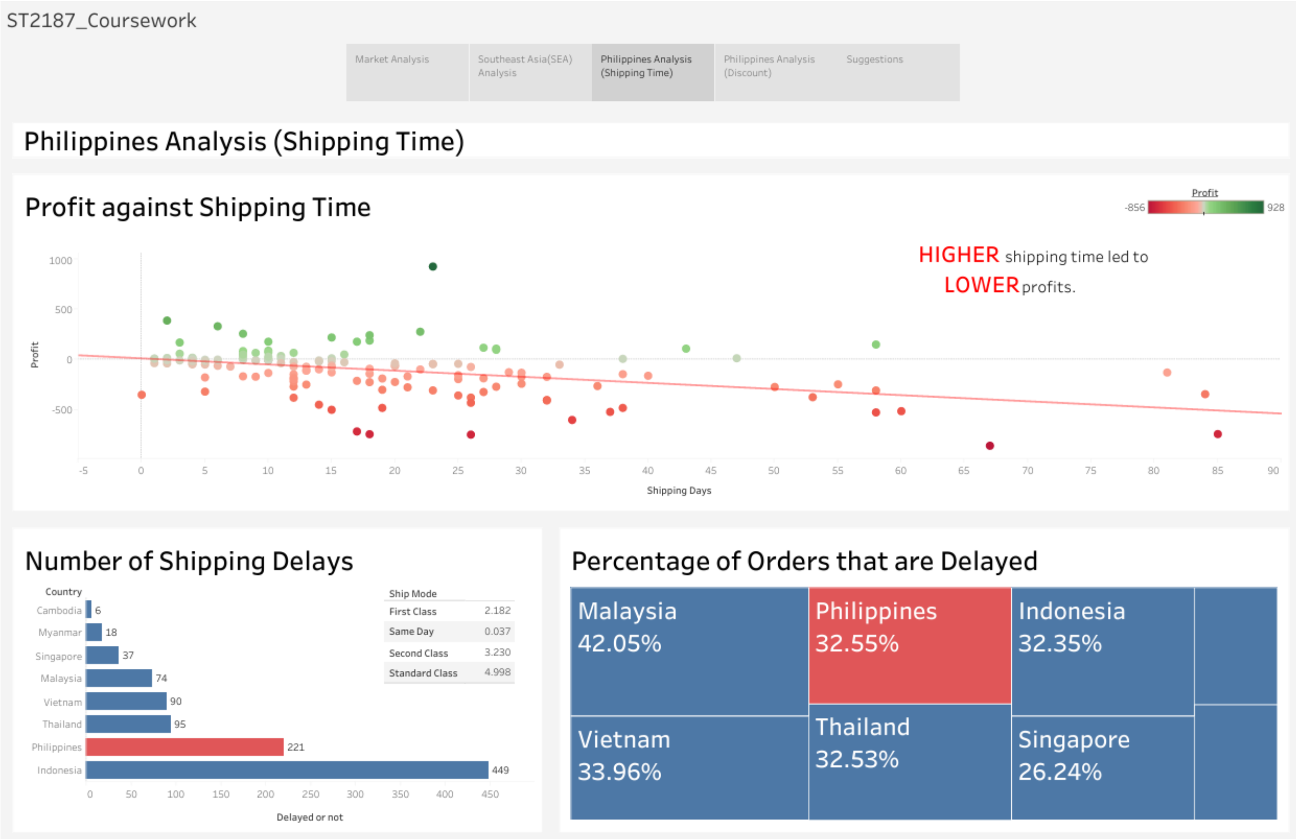


Fig 8. Philippines Analysis on Shipping time dashboard

Moving on to the next dashboard (Figure 5), we look at how shipping issues are costing Philippines problem.

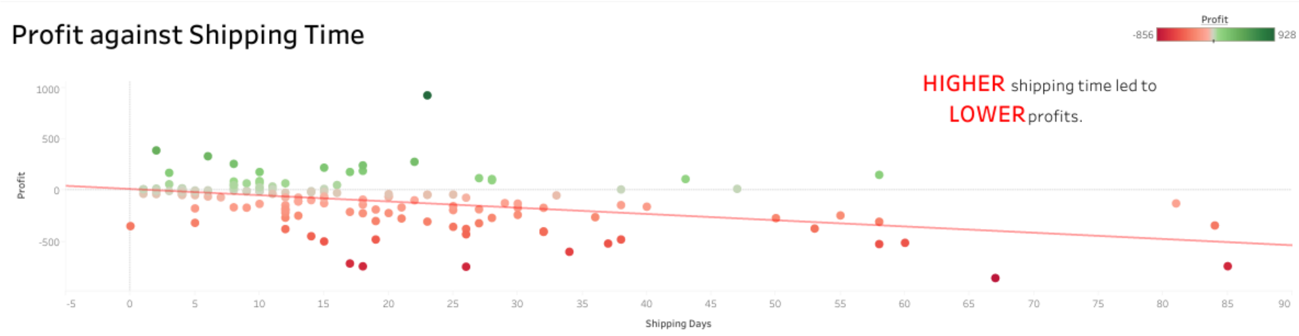


Fig 9. Profit against shipping time

The analysis has shown us that there's a negative correlation between shipping days and profit. This means that as shipping time increases, profits generally decrease across the board. With this in mind, we should try to priorities minimizing shipping time to enhance profitability of Philippines.

Ship Mode	
First Class	2.182
Same Day	0.037
Second Class	3.230
Standard Class	4.998

Fig 10. Average shipping days for each shipping mode

Shipping delays were determined by comparing the actual shipping times against the average shipping time (Figure 10) of the different shipping modes. The benchmarks were as followed:

- First Class: 2 days
- Same Day: 0 days
- Second Class: 3 days
- Standard Class: 5 days

Any shipment exceeding the average was deemed as delayed. This enabled us to find the amount of shipping delays and conduct further analysis on shipping delays.

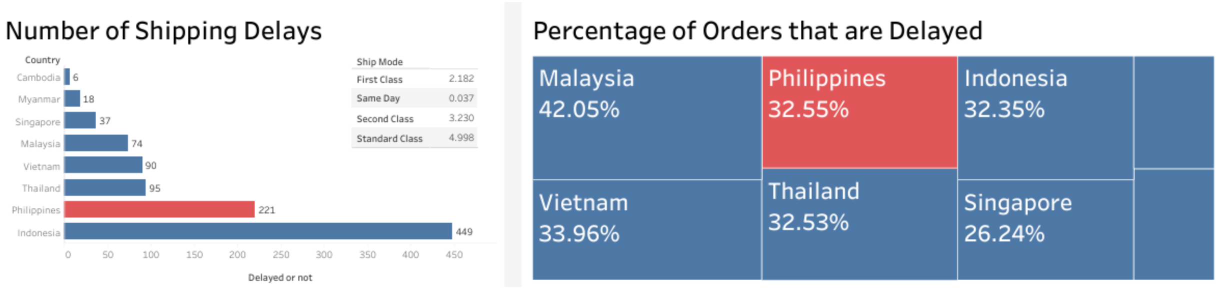
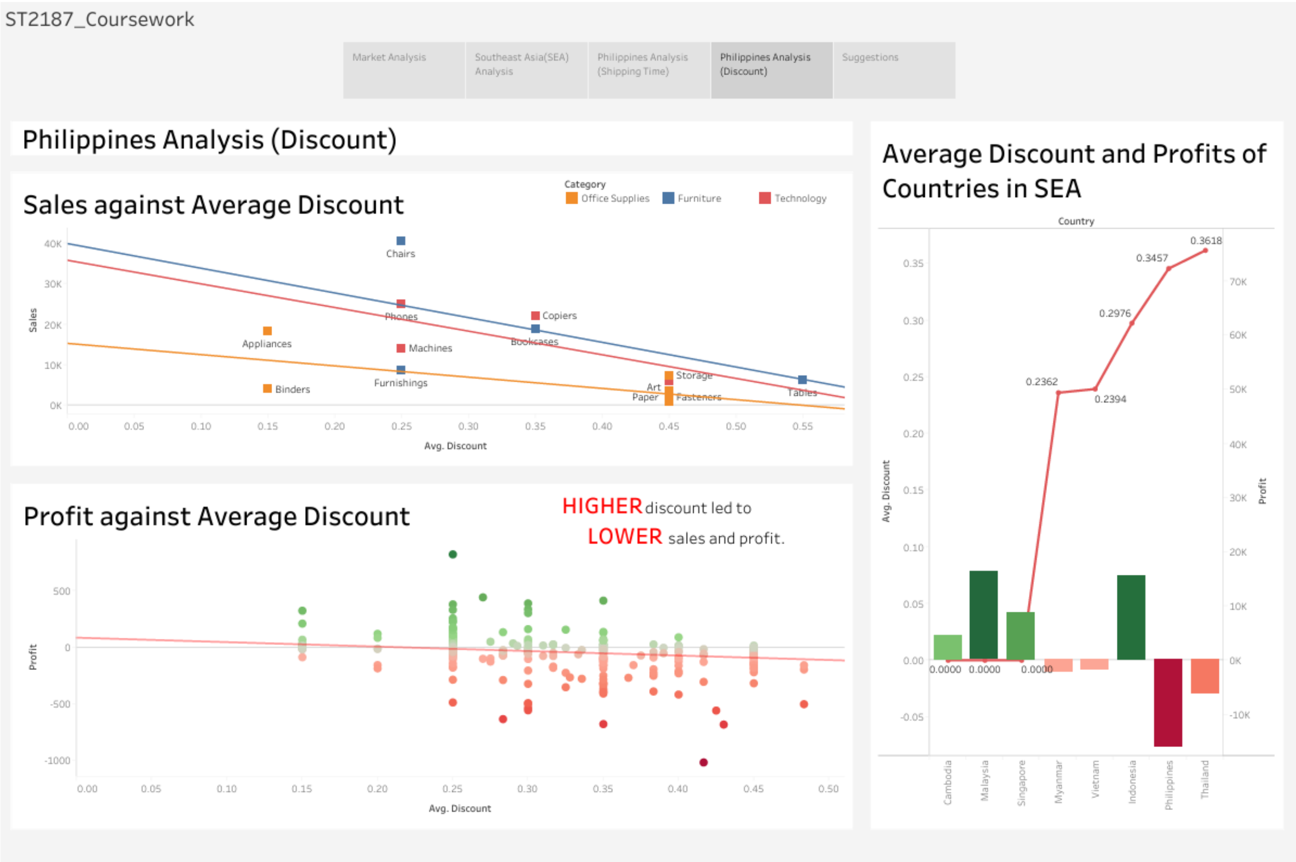


Fig 11. Number of shipping delays and Percentage of orders that are delayed

The Philippines has the second highest number of shipping delays in Southeast Asia, which may be due to its relatively higher orders placed. However, when considering the percentage of delayed orders, the Philippines is ranked third, with 32.55% of orders experiencing some kind of delay.

These delays negatively affect customer satisfaction, leading to a lower repeat customers and lower sales in the future should these customers leave bad reviews. This shows the importance of delivering on time. Hence, addressing these shipping delays should be a priority to prevent future losses and improve the business.

2.4. Dashboard 4: Philippines Analysis (Discount)



<u>Fig 11. Philippines analysis on discount dashboard</u>

To find out how improper usage of discounts affect profits, we created this dashboard (Figure 11). This dashboard will find out whether higher discounts will lead to increase in sales and what is the impact of issuing high discounts. By identifying the trends and potential issues, we will be able to optimize discount strategies to maximize profits instead of eating into the profits.

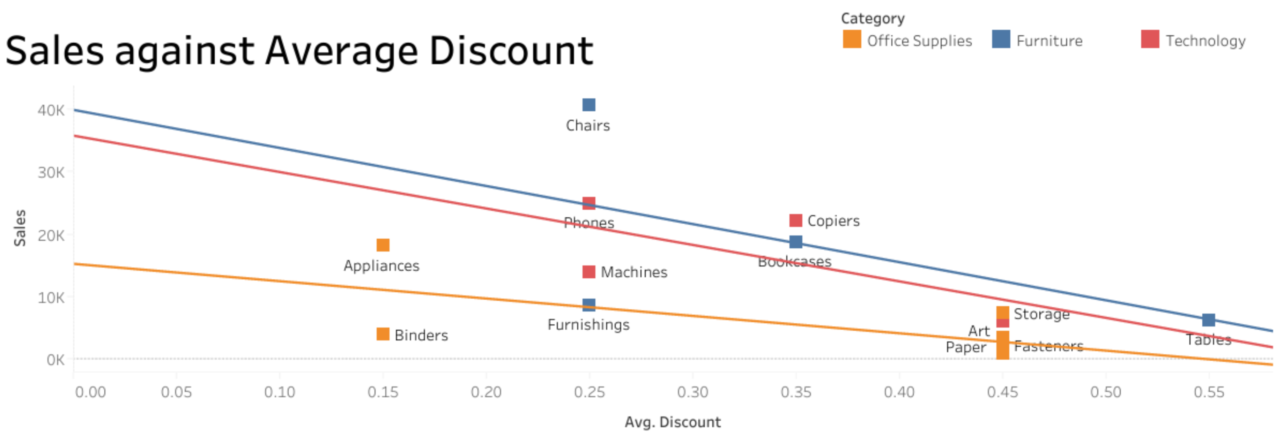


Fig 12. Sales against Average discount scatterplot

The scatterplot (Figure 12) shows the relationship between sales and average discount. In this case, there are negative correlation between sales and average discount. The negative trend lines for the different categories indicate that as average discount increases, total sales decrease.

This means that in general, higher discounts does not lead to higher sales. This would lead to the business needing to reevaluate and adjust its discount strategies to optimize sales and profits.

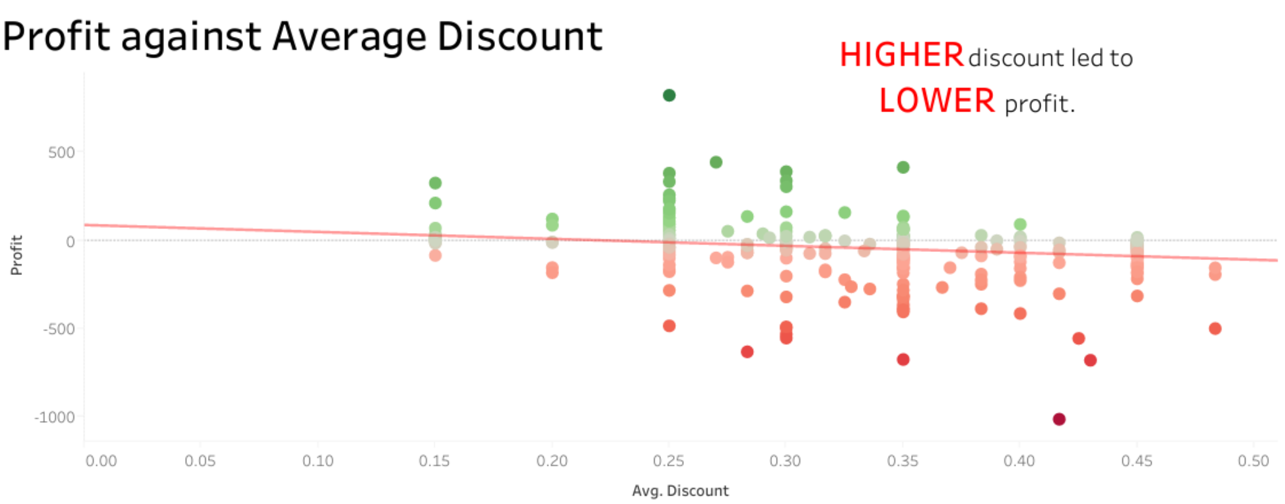


Fig 13. Profit against Average discount scatterplot

The scatterplot (Figure 13) shows the relationship between profit and average discount. In this case, there are negative correlation between profit and average discount. The negative trend lines indicates that as average discount increases, profit decrease.

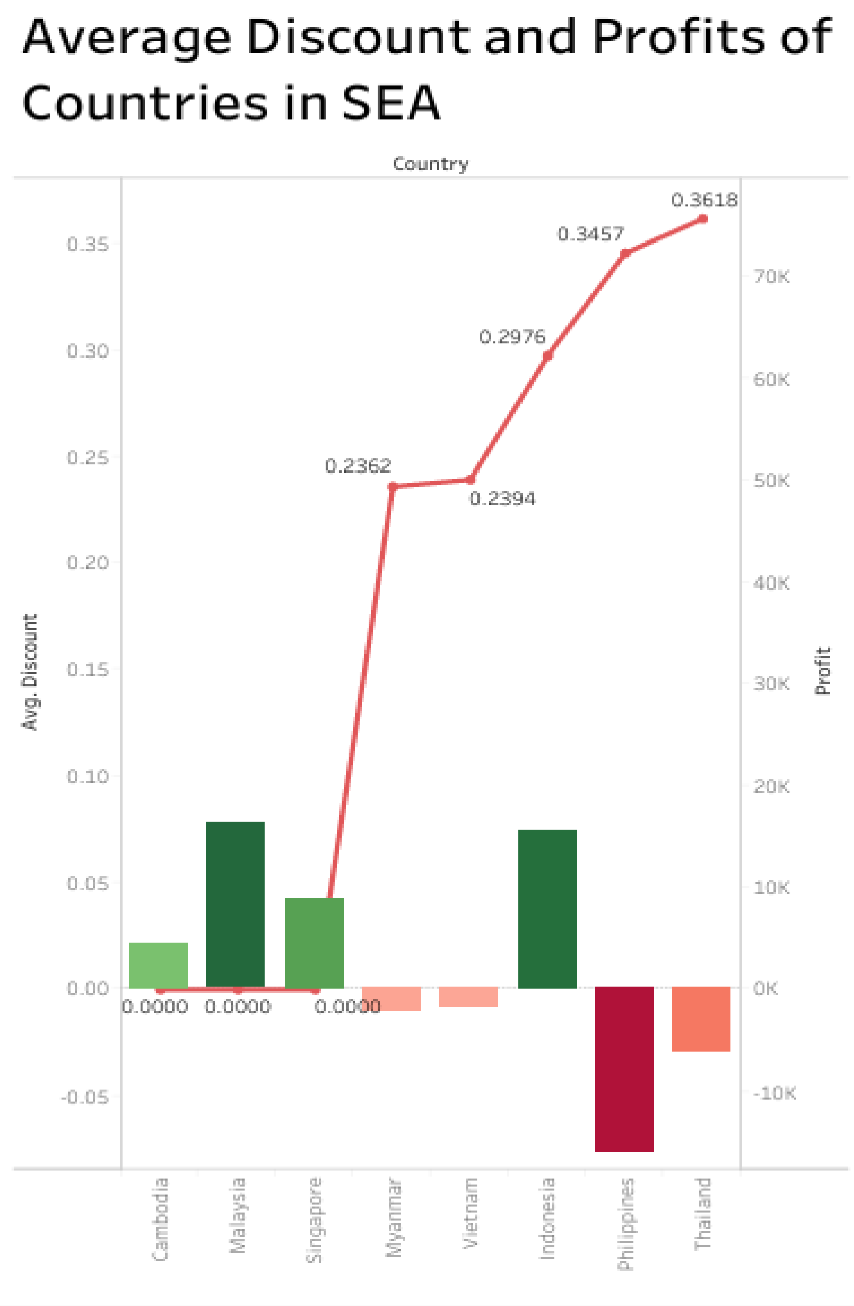


Fig 14. Average discounts and Profits of countries in SEA

In Southeast Asia, Philippines has the second highest average discount applied to its items sold. In general, the countries that offers higher discount tend to have lower profits as demonstrated by Philippines and Thailand. These two countries, which provided the highest discounts, were the two countries that recorded the lowest profits.

This means that excessive discounting negatively impacts profitability. A reason for that may be that the high discount provided is reducing profit margins. Therefore, discount strategies should be carefully managed and applied in a controlled manner to optimize financial performance.

3. Summarised findings from analysis

3.1. Lowest performing region and country

With a profit margin of just 2.15%, Southeast Asia is underperforming and in it, Philippines has the lowest profit margin.

3.2. Shipping delays causing loss

Longer shipping times in the Philippines negatively affects profitability, making it important to minimize delay. With the second highest number of shipping delays in Southeast Asia and 32.55% of orders affected by shipping delays, improvements must be made to improve customer satisfaction.

3.3. Discounts given are too high

There is a negative correlation between average discount against profit and sales. To maximise profits, discount strategies must be carefully managed and applied in controlled amounts rather than relying on excessive discounts.

With these findings, we will now move on to give our recommendations in the final dashboard.

4. Recommendations and targets

4.1. Dashboard 5: Suggested solutions

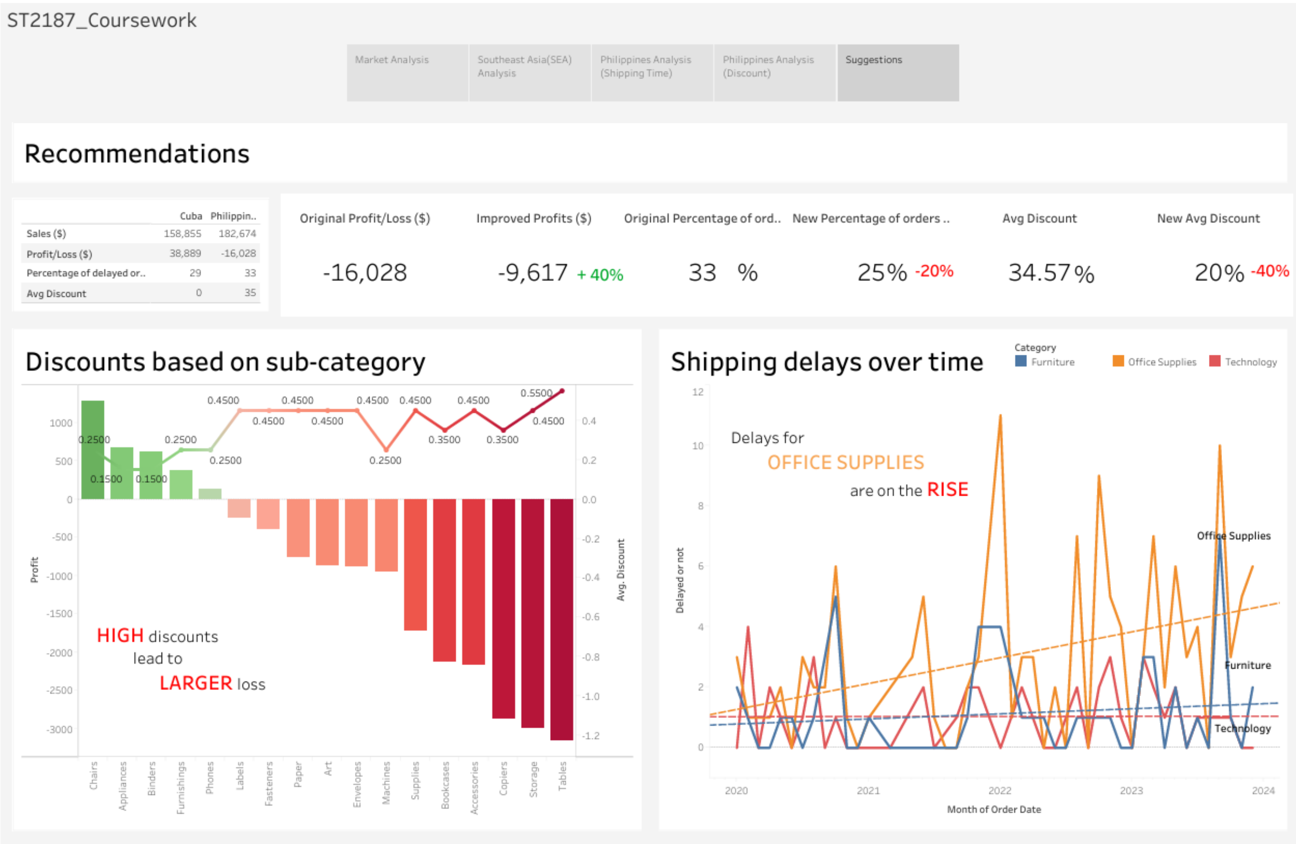


Fig 15. Recommendations dashboard

Based on insights from Dashboard 5 (Figure 15), we recommend two key strategies to enhance the business's profitability and operational efficiency.

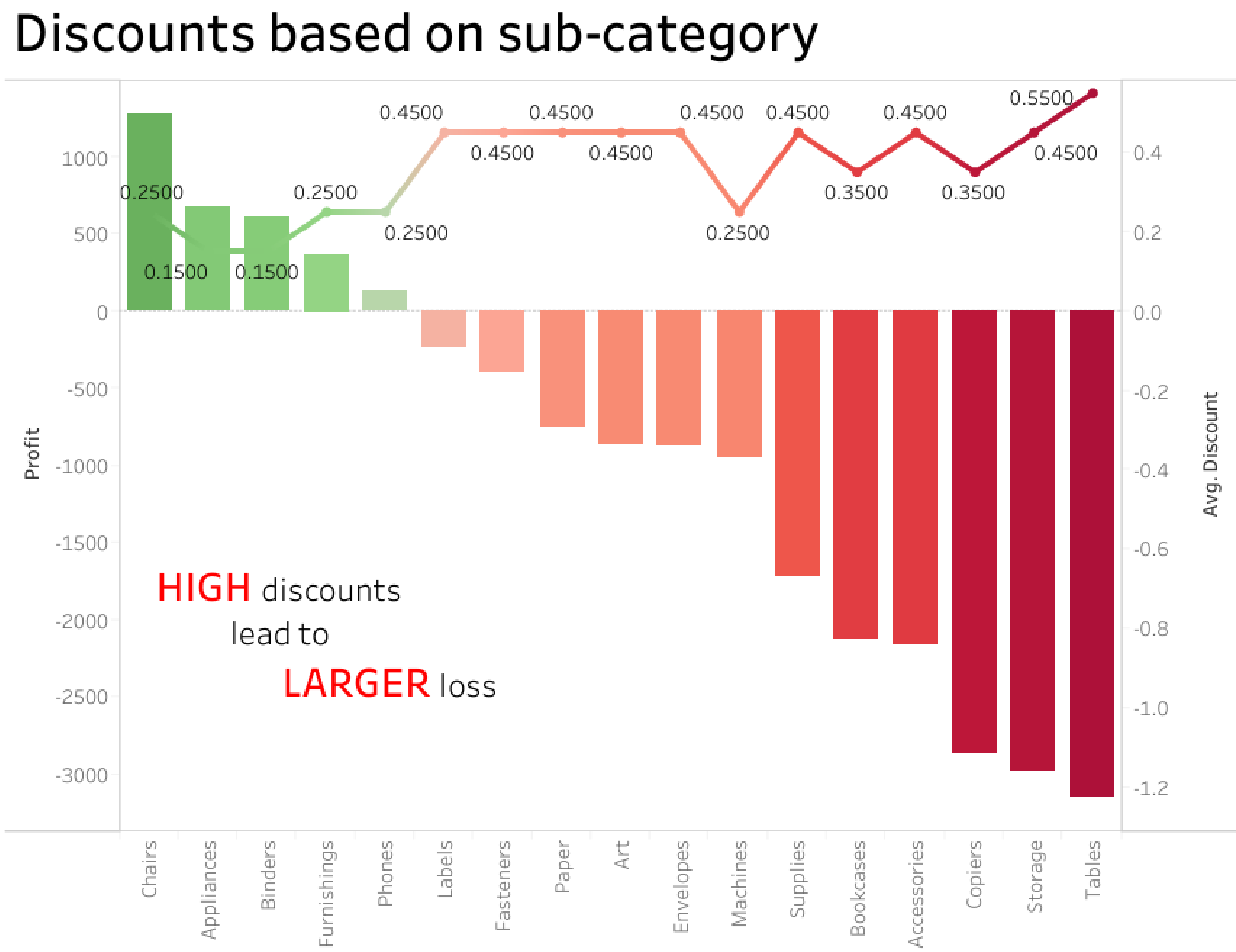


Fig 16. Discount given based on sub-category and its profits

First, the business should consider reducing discounts for the sub-category in red. Instead of solely relying on high discounts to drive sales, an alternative solution would be to implement targeted marketing campaigns. The business should still use discounts to promote sales but at a lower rate so that profits are not being negatively impacted. By optimizing discount structures, the business can increase profitability.

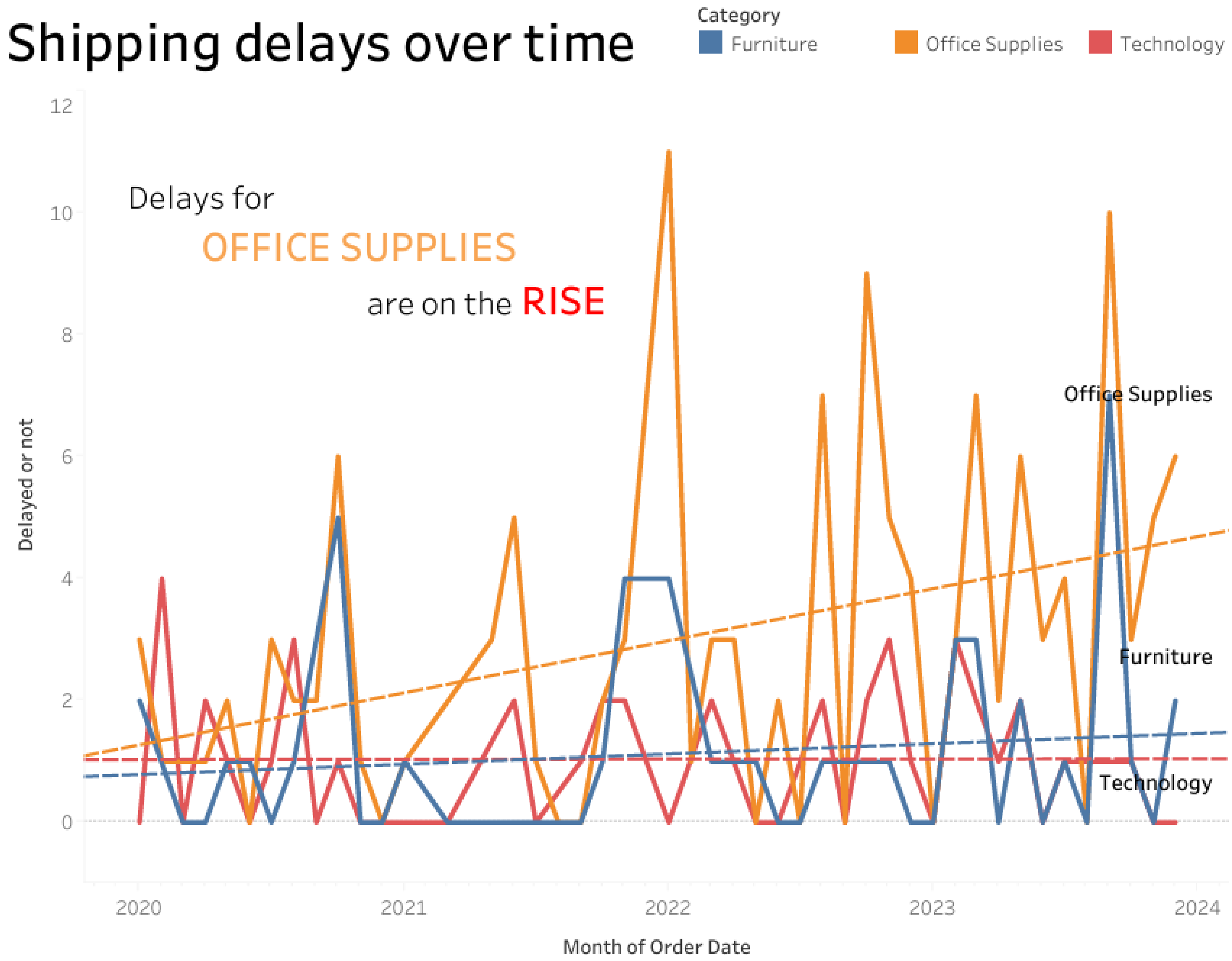


Fig 17. Shipping delays of categories over time

Secondly, it is essential to make sure to deliver these office supplies on time. The graph (Figure 17) shows that delays for office supplies has been increasing. Since many of these orders come from businesses, consistent delays could negatively impact their operations, causing they to not want to purchase from us again. Addressing this issue would help maintain a good relationship with these businesses and help maintain sales.

	Cuba	Philippines
Sales ($)	158,855	182,674
Profit/Loss ($)	38,889	-16,028
Original Percentage of orders delayed	29	33
Original Avg Discount	0	35

Fig 18. Reference country

For reference, Cuba was chosen as it has almost identical sales figure as Philippines. However, they were able to generate better profits at $38,889 compared to Philippines's loss of $16,028. Both their percentage of delayed orders and average discount is much lower than Philippines which makes them a good reference country.

Original Profit/Loss ($)	Improved Profits ($)	Original Percentage of ord..	New Percentage of orders ..	Original Avg Discount	New Avg Discount
-16,028	-9,617 + 40%	32.55 %	25% -20%	34.57 %	20% -40%

Fig 19. Target goals for the business

This (Figure 19) are the goals that we have set for the company. By achieving a 20% reduction in delayed orders and lowering the average discount rate to 40%, we anticipate a 40% increase in profits. If this strategy proves successful, the business may consider applying it to other underperforming markets.

Conclusion

With the rapidly increasing market, Southeast Asia is an important region for the business to take note of. To thrive in this new business opportunity, the business needs to lower discounts and reduce shipping delays. By addressing these issues, the business can expect a projected 40% increase in profits over the next year for the Philippines. Additionally, if these strategies prove useful, they can be used on other underperforming markets, further enhancing business growth and profits.

References

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